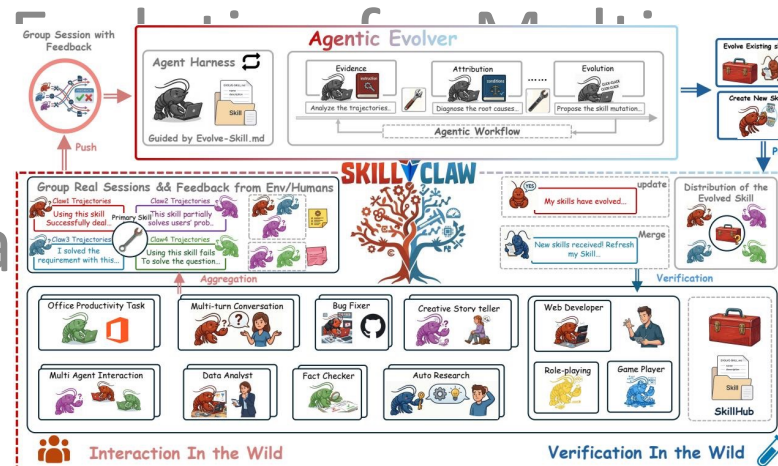


SkillClaw: Let Skills Evolve Collectively with Agentic Evolver

Collective Skill
User LLM Agent
[arxiv.org/a](https://arxiv.org/abs/2406.11101)



Agent Skills Remain Static After Deployment — Users Rediscover

Solutions Independently

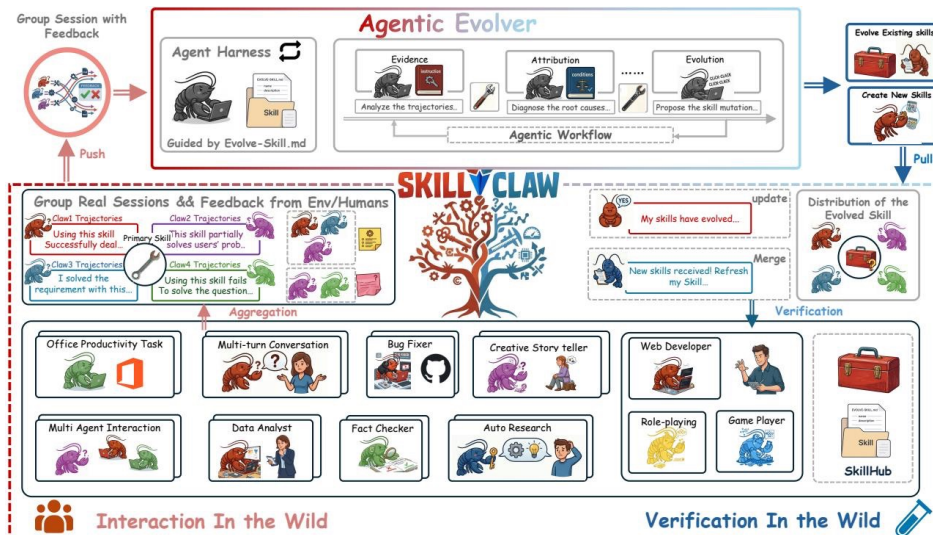
- The Core Problem: LLM agent skills do not evolve post-deployment
- **Static Skill Paradigms:**
 - Skills are manually installed from centralized hubs.
 - Solutions discovered during interaction rarely persist beyond individual sessions.
 - Same patterns of failure and recovery repeatedly observed.
- **The Consequence:**
 - Each user rediscovers solutions independently.

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill

Multi-user interaction → Session collection → **Evolution** → Agentic evolver → Skill synchronization

The Four Stages:

1. Evidence: Analyze interaction trajectories to identify behavioral patterns.
2. Attribution: Diagnose root causes of failures and successes.
3. Evolution: Propose skill mutations (refine or create new skills).
4. Distribution/Synchronization: Verified skills merged to shared repo and synced to agents.



Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

- Three-Stage Agentic Workflow:
- **1. Evidence Analysis:**
 - Analyze aggregated trajectories to identify recurring behavioral patterns.
- **2. Attribution Diagnosis:**
 - Diagnose the root causes of both failures and successes in the evidence base.
- **3. Evolution & Mutation:**
 - Propose skill mutations, refining existing skills or creating new ones.

Cross-User Trajectory Aggregation

Exposes Skill Behavioral Boundaries

- The Value of Cross-User Aggregation:
- **Compatible Claw-style systems:**
 - OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw.
- **Diverse contexts produce complementary views of a skill's behavioral boundary:**
 - Reveals both conditions under which a skill works and those under which it breaks.
- **Single-user limitations:**
 - A single user rarely generates enough signal to

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

Day	Candidate Skill	Skill Function	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Check inputs & env setup	Accept	Upgrade best pool
2	multimodal-input-validation-and-setup	Validation & env init	Reject	Keep current best
3	multimodal-creative-task-pipeline	Creative pipeline (extract...)	Reject	Keep current best
4	multimodal-creative-task-pipeline (impr.)	Add image classification	Reject	Keep current best
5	multimodal-creative-task-pipeline... + validate...	Pipeline + per-file fail-fast	Reject	Keep current best
6	multimodal-creative-task-pipeline (cand.)	Extend PDF-to-poster paths	Reject	Not admitted

Key Finding: Only 1 of 6 candidates was accepted, illustrating the conservative verification process that prevents regressions.

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Day	Candidate Skill	Change Summary	Validator	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking	Accept	Add to Safety pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames	Accept	Keep updated pool
3	git-push-with-auth-fallback + git-clone...	Auth-alternative paths & audit	Accept	Keep current best
4	(none)	Same-pool retest; confirm best	Accept	Continue best pool
5	git-push-with-auth-fallback	Add 'push hang treated as auth fail'	Reject	Keep current best
6	git-push-with-auth-fallback	Add identity config & consistency	Reject	Not admitted

Key Finding: 3 of 6 days yielded accepted updates, showing successful iterative refinement before reaching a plateau.

Key Takeaways: Three Properties That Distinguish SkillClaw

- **Three Distinguishing Properties:**
- **1. Collective Evolution:**
 - Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem.
- **2. Fully Automatic:**
 - Skill evolution is driven by runtime interaction without manual curation or explicit user intervention.
- **3. Agentic Evolution Paradigm:**